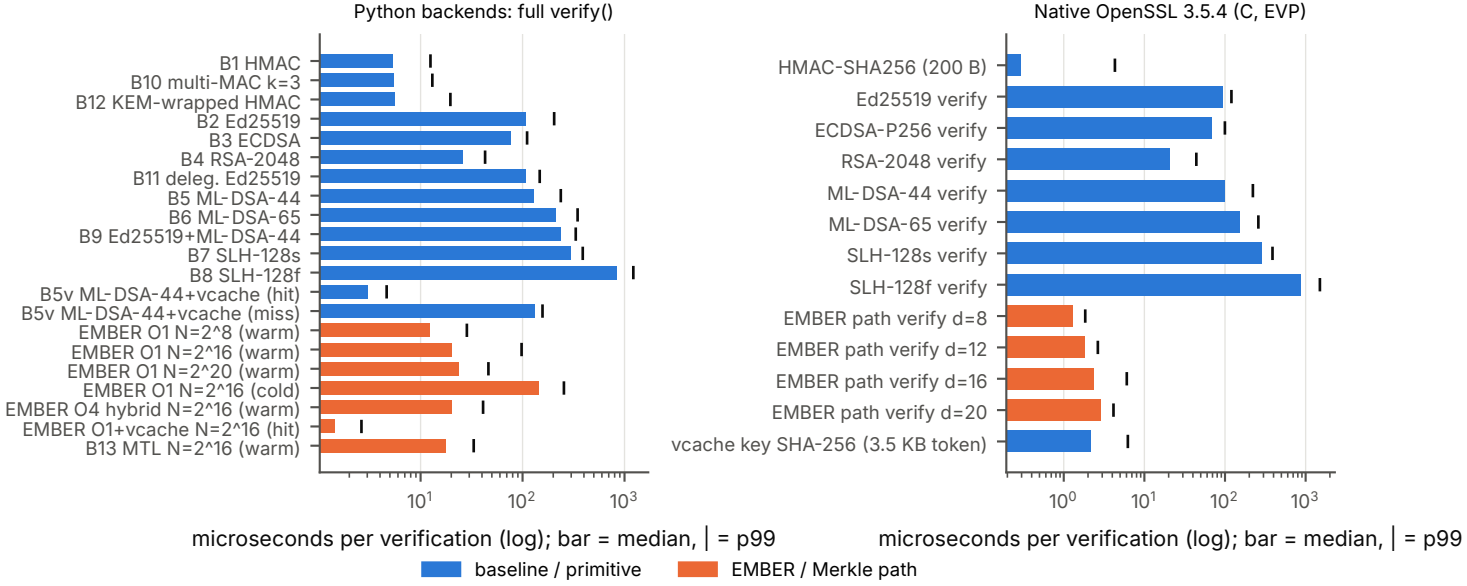
Python backends: full verify()
B1 HMAC
B10 multi-MAC k=3
B12 KEM-wrapped HMAC
B2 Ed25519
B3 ECDSA
B4 RSA-2048
B11 deleg. Ed25519
B5 ML-DSA-44
B6 ML-DSA-65
B9 Ed25519+ML-DSA-44
B7 SLH-128s
B8 SLH-128f
B5v ML-DSA-44+vcache (hit)
B5v ML-DSA-44+vcache (miss)
EMBER O1 N=2^8 (warm)
EMBER O1 N=2^16 (warm)
EMBER O1 N=2^20 (warm)
EMBER O1 N=2^16 (cold)
EMBER O4 hybrid N=2^16 (warm)
EMBER O1+vcache N=2^16 (hit)
B13 MTL N=2^16 (warm)
10^1
10^2
10^3
microseconds per verification (log); bar = median, | = p99
Native OpenSSL 3.5.4 (C, EVP)
HMAC-SHA256 (200 B)
Ed25519 verify
ECDSA-P256 verify
RSA-2048 verify
ML-DSA-44 verify
ML-DSA-65 verify
SLH-128s verify
SLH-128f verify
EMBER path verify d=8
EMBER path verify d=12
EMBER path verify d=16
EMBER path verify d=20
vcache key SHA-256 (3.5 KB token)
10^0
10^1
10^2
10^3
microseconds per verification (log); bar = median, | = p99
baseline / primitive
EMBER / Merkle path